ICC MEN'S T20 WORLD CUP 2022

Analytics Intelligence Dashboard

Comprehensive Project Report

> *A data-driven visual intelligence platform engineered for cricket selectors, coaches,*
> *and performance analysts — transforming raw match data into strategic squad decisions.*

Prepared By	Domain	Tool Stack	Report Date
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42 Matches | 16 Teams | 7 Venues | 213 Players | 515 Wickets

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1. Executive Summary

The ICC Men's T20 World Cup 2022, hosted across seven venues in Australia, served as one of the most analytically rich cricket tournaments of the modern era. Spanning 42 matches, 16 participating nations, 213 unique players, and 515 total wickets, the tournament generated a substantial volume of granular, match-level performance data — data that, until now, has been consumed primarily through traditional scorecard formats that offer limited strategic utility.

This project delivers a production-grade Power BI Analytics Intelligence Dashboard that systematically transforms raw batting, bowling, and match-summary records into a structured, six-page visual intelligence platform. Designed specifically for cricket selectors, head coaches, and team analysts, the dashboard provides a clear, data-backed framework for answering the most consequential questions in tournament cricket: who performs under pressure, who exploits specific match conditions, and who delivers consistent returns across an extended tournament schedule.

Total Matches	Teams	Players Profiled	Total Runs	Wickets Taken	Venues Covered
42 All stages	**16** International	**213** Bat & Bowl	**11K+** Tournament	**515** All formats	**7** Australia

The dashboard is structured across six purpose-built analytical pages — Overview, Openers, Middle Order, All Rounders, Bowlers, and Stats — each leveraging a bespoke set of DAX measures and a Star Schema data model to deliver role-specific, filter-responsive insights. Stage-level slicers (Group Stage, Super 12, Playoffs) allow users to isolate performance under progressively high-stakes conditions, offering the most nuanced view of player reliability available from the public tournament dataset.

The primary conclusion drawn from this analysis is that data-driven squad selection — grounded in consistency metrics, venue-adjusted statistics, and match-up pattern recognition — provides a measurable competitive advantage over conventional selection approaches. The recommendations presented in Section 6 offer a concrete, actionable framework for selectors seeking to apply these findings to future T20 tournament strategy.

2. Problem Statement & Business Objectives

2.1 The Core Problem

Cricket selection has historically been governed by a combination of expert intuition, recent form observation, and reputational precedent. While these inputs carry contextual value, they are inherently subjective and often fail to account for the multidimensional nature of modern T20 performance. A batsman averaging 38 across five innings may look statistically strong, yet that average conceals critical variance — whether those runs came against weaker group-stage opposition, in favourable chase conditions, or against specific bowling archetypes that will not be present in knockout rounds.

The absence of an integrated analytical platform means selectors are forced to triangulate between disparate scorecard data, subjective scouting reports, and anecdotal match memories. This produces selection committees that are information-rich but insight-poor — possessing raw data without the analytical infrastructure to convert it into reliable, role-specific conclusions.

2.2 Business Objectives

This project was initiated to address six core business objectives:

- **Objective 1 — Centralise Performance Data:** Consolidate batting, bowling, and match metadata into a single, queryable analytical environment accessible to all members of the selection committee.
- **Objective 2 — Enable Role-Specific Analysis:** Move beyond aggregate team statistics to provide granular, role-filtered insights for Openers, Middle Order, All Rounders, and Bowlers independently.
- **Objective 3 — Quantify Consistency:** Introduce statistical measures — particularly the Consistency Index (Average Runs divided by Standard Deviation, scaled to 100) — that capture performance reliability beyond simple averages.
- **Objective 4 — Stage-Stratified Intelligence:** Allow selectors to filter all metrics by tournament stage (Group Stage, Super 12, Playoffs) to understand how player performance evolves under escalating competitive pressure.
- **Objective 5 — Venue & Condition Profiling:** Map runs, wickets, and boundary percentages across all seven Australian venues to identify pitch-type advantages and ground-specific player performance patterns.

- **Objective 6 — Strategic Recommendations:** Translate dashboard insights into concrete squad construction recommendations, including batting-order optimisation, bowling attack composition, and All Rounder selection criteria.

2.3 Scope & Limitations

The analytical scope of this project is confined to the ICC T20 World Cup 2022 tournament dataset. The following limitations are acknowledged:

- Player data reflects tournament performance exclusively; career averages and prior T20I history are not incorporated into this model.
- The NRR (Net Run Rate) calculation is computed from within-dataset batting and bowling ball totals, which may produce marginal rounding differences relative to official ICC figures.
- Player image data referenced in the Players dimension table is metadata-linked and not rendered within the analytical pages of this report.
- Match-up analysis (individual batter vs. specific bowler type) is not available at ball-by-ball resolution within the current dataset schema.

3. Data Understanding & Preparation

3.1 Raw Data Sources

The project dataset comprises four inter-related CSV files sourced from publicly available ICC T20 World Cup 2022 records and supplemented with curated player profile data. Each file targets a distinct analytical domain:

Dataset	Primary Key	Records	Key Fields
batting_summary.csv	match_id + batsmanName	~1,800 rows	runs, balls, fours, sixes, out, battingPos
bowling_summary.csv	match_id + bowlerName	~1,400 rows	balls, wickets, runs, economy, maiden, zeros
match_summary.csv	match_id	42 rows	team1, team2, ground, matchDate, margin, winner, Stage
players.csv	name	213 rows	team, battingStyle, bowlingStyle, playingRole, image

3.2 Python Data Cleaning Pipeline

All raw CSV files were processed through a Python-based data preparation pipeline implemented in Pandas and NumPy before ingestion into Power BI. The cleaning steps were systematically applied in the following sequence:

3.2.1 Library Imports & File Loading

```
import pandas as pd
import numpy as np

bat  = pd.read_csv('batting_summary.csv')
bowl = pd.read_csv('bowling_summary.csv')
match = pd.read_csv('match_summary.csv')
players = pd.read_csv('players.csv')
```

3.2.2 Data Type Enforcement

Numeric columns stored as strings (common in scorecard exports) were explicitly cast to integer or float types. The 'out' column — representing dismissal status — was converted from text ('out'/'not out') to a binary integer flag (1/0):

```
bat['runs']  = pd.to_numeric(bat['runs'],  errors='coerce').fillna(0).astype(int)
bat['balls'] = pd.to_numeric(bat['balls'], errors='coerce').fillna(0).astype(int)
bat['fours'] = pd.to_numeric(bat['fours'], errors='coerce').fillna(0).astype(int)
bat['sixes'] = pd.to_numeric(bat['sixes'], errors='coerce').fillna(0).astype(int)
bat['out']   = bat['out'].apply(lambda x: 1 if str(x).strip().lower() == 'out' else
0)

bowl['balls']   = pd.to_numeric(bowl['balls'],
errors='coerce').fillna(0).astype(int)
bowl['wickets'] = pd.to_numeric(bowl['wickets'],
errors='coerce').fillna(0).astype(int)
bowl['runs']    = pd.to_numeric(bowl['runs'],
errors='coerce').fillna(0).astype(int)
bowl['economy'] = pd.to_numeric(bowl['economy'], errors='coerce').fillna(0.0)
```

3.2.3 Missing Value Treatment

Missing values were identified and treated according to field-specific logic:

- **Numeric fields (runs, balls, wickets):** Null values filled with 0 using fillna(0), as missing match entries represent non-participation rather than data corruption.
- **String fields (batsmanName, bowlerName):** Rows with null name fields were dropped entirely as they cannot be attributed to any player profile.
- **match_summary[margin]:** Null margins (no-result matches) were retained with a null flag; the Win Type calculated column handles this gracefully via CONTAINSSTRING logic.
- **players[image]:** Null image URLs were retained as metadata; they do not impact any KPI calculation.

3.2.4 Feature Engineering

The following derived fields were created during the Python preparation phase to support downstream DAX calculations:

```
# Boundary run contribution (pre-computed as a validation check)
bat['boundary_runs_py'] = bat['fours'] * 4 + bat['sixes'] * 6
```

```
# Match stage normalisation (standardise text casing)
match['Stage'] = match['Stage'].str.strip().str.title()

# Date parsing
match['matchDate'] = pd.to_datetime(match['matchDate'], dayfirst=True)

# Economy rate recompute (verify against raw ball count)
bowl['economy_check'] = (bowl['runs'] / (bowl['balls'] / 6)).round(2)

# Export cleaned files
bat.to_csv('batting_clean.csv',   index=False)
bowl.to_csv('bowling_clean.csv',  index=False)
match.to_csv('match_clean.csv',   index=False)
players.to_csv('players_clean.csv', index=False)
```

3.3 Data Quality Summary

Following the cleaning pipeline, the dataset achieved the following quality benchmarks:

Quality Dimension	Batting	Bowling	Match	Players
Null Records Removed	< 0.3%	< 0.2%	0%	0%
Type Consistency	100%	100%	100%	100%
Duplicate Rows	None	None	None	None
Referential Integrity	Verified	Verified	N/A	N/A

4. Data Modeling & DAX Formulation

4.1 Schema Architecture — Star Schema

The Power BI data model adopts a Star Schema architecture — the industry-standard approach for analytical workloads. Two fact tables sit at the centre of the model, each connected to two shared dimension tables via clearly defined one-to-many relationships. This design ensures:

- Optimal DAX filter propagation from dimension to fact tables
- Minimal data redundancy across the model
- Efficient cross-filtering between batting and bowling domains
- Scalable extension for future tournament datasets

4.2 Table Classification

Table	Type	Granularity	Row Count
bating_summary	FACT	One row per player per match innings	~1,800
bowling_summary	FACT	One row per bowler per match innings	~1,400
match_summary	DIMENSION	One row per match	42
players	DIMENSION	One row per player	213

4.3 Relationship Map

The four tables are connected through the following relationships, all configured as single-directional with referential integrity enforced:

From Table	From Column	To Table	To Column	Cardinality
match_summary	match_id	bating_summary	match_id	1 : Many
match_summary	match_id	bowling_summary	match_id	1 : Many
players	name	bating_summary	batsmanName	1 : Many
players	name	bowling_summary	bowlerName	1 : Many

4.4 Calculated Columns

Three calculated columns were created within Power BI to extend the base tables with derived attributes that are evaluated row-by-row at data load time:

4.4.1 Boundary Runs [bating_summary]

```
Boundary runs = bating_summary[fours]*4 + bating_summary[sixes]*6
```

Logic: A four scores 4 runs; a six scores 6 runs. This column isolates boundary-exclusive run contribution per innings and is the denominator input for the Boundary % KPI measure.

4.4.2 Opponent [bating_summary]

```
Opponent =
VAR CurrentTeam = RELATED('players'[team])
VAR MatchName   = 'bating_summary'[match]
VAR RemoveTeam  = SUBSTITUTE(MatchName, CurrentTeam, "")
VAR CleanOpp    = SUBSTITUTE(RemoveTeam, "Vs", "")
RETURN TRIM(CleanOpp)
```

Logic: The match field stores a composite string in the format 'Team1 Vs Team2'. By fetching the current player's team via RELATED and substituting it out of the match string (along with the 'Vs' separator), the remaining text resolves to the opponent team name.

4.4.3 Win Type [match_summary]

```
Win Type =
IF(
    CONTAINSSTRING(match_summary[margin], "runs"),
    "Defending",
    "Chasing"
)
```

Logic: In cricket, a victory margin expressed in 'runs' means the first-batting team won (Defending). A margin expressed in 'wickets' means the second-batting team won (Chasing). This classification supports match-type segmentation in the Overview analysis.

4.5 Key DAX Measures

All measures are organised into four logical groups: Batting, Bowling, Point Table, and Others. The following tables summarise the mathematical intent behind each measure:

4.5.1 Batting Measures

Measure	Formula Logic	Business Purpose
Total Runs	SUM(bating_summary[runs])	Aggregate run production for filtered context
Average Runs	AVERAGE(bating_summary[runs])	Mean runs per innings — baseline performance level
Batting Average	DIVIDE([Total Runs],[Total Innings Dismissed],0)	Runs per dismissal — standard quality benchmark
Strike Rate	DIVIDE([Total Runs],[Total Ball Faced],0)*100	Run scoring efficiency per 100 balls
Boundary %	DIVIDE(SUM([Boundary runs]),[Total Runs],0)*100	% of total runs from boundaries — aggressive intent
Consistency Index	ROUND(DIVIDE([Avg Runs],[StdDev Runs],0)*100,2)	Reliability score; higher = less variance per innings
Highest Indiv. Score	MAXX(ALLSELECTED(bating_summary),[runs])	Peak single-innings score in filtered context
Running Btwn Wickets	SUM([runs]) - ([Runs from 4s]+[Runs from 6s])	Runs scored by running — fitness and placement proxy

4.5.2 Bowling Measures

Measure	Formula Logic	Business Purpose
Wickets	SUM(bowling_summary[wickets])	Total dismissals — match-winning impact
Bowling Average	DIVIDE([Run Conceded],[Wickets],0)	Runs per wicket — efficiency of wicket-taking
Bowling Economy	DIVIDE([Run Conceded],([Balls Bowled]/6),0)	Runs per over — containment effectiveness
Bowling Strike Rate	DIVIDE([Balls Bowled],[Wickets],0)	Balls per wicket — wicket-taking frequency
Dot Ball %	DIVIDE(SUM([zeros]),SUM([balls]),0)*100	Pressure-building metric — % deliveries scoring zero
Top 5 Wickets	CALCULATE([Wickets],KEEPFILTERS(TOPN(5,...)))	Dynamic visual measure for Top 5 ranking charts

4.5.3 Point Table Measures

The Point Table measures represent the most complex DAX engineering in the project. They dynamically compute each team's match count, wins, losses, points, and NRR within the current filter context — enabling a live, interactive standings table without a pre-aggregated source table.

```
-- Net Run Rate (most complex measure)
NRR =
VAR CurrentTeam   = SELECTEDVALUE(players[team])
VAR RunsScored    = CALCULATE(SUM(bating_summary[runs]),  bating_summary[team] =
CurrentTeam)
VAR OversFaced    = CALCULATE(SUM(bating_summary[balls]),  bating_summary[team] =
CurrentTeam) / 6
VAR RunsConceded  = CALCULATE(SUM(bowling_summary[runs]), bowling_summary[Team] =
CurrentTeam)
VAR OversBowled   = CALCULATE(SUM(bowling_summary[balls]),bowling_summary[Team] =
CurrentTeam) / 6
RETURN
DIVIDE(RunsScored, OversFaced) - DIVIDE(RunsConceded, OversBowled)
```

5. Visual Analysis & Dashboard Walkthrough

The dashboard is architecturally designed around a consistent three-zone page layout: a Header Zone (Logo, Navigation Buttons, Stage Slicer), a KPI Ribbon (six metric cards), and a Charts Zone (primary analytical visuals and a detail table). This structure ensures that every page delivers immediate high-level context before demanding detailed interpretation from the user.

5.1 Page 1 — Tournament Overview

Purpose

The Overview page functions as the executive command centre of the dashboard. It provides a complete tournament-level summary and is designed to be interpretable by non-technical stakeholders within seconds of opening. It answers the question: 'How did the tournament play out at a macro level?'

KPI Ribbon

Players	Teams	Venues	Matches	Total Runs	Total Wickets
213	16	7	42	11K	515

Visual Design Rationale

- **Point Table (Matrix Visual):** A matrix was chosen over a standard table to enable conditional formatting on the NRR column — green backgrounds for positive NRR, red for negative — giving selectors an immediate visual gradient of team competitiveness. England (10 pts, NRR +0.18) led the table, with New Zealand (+1.05 NRR) showing the tournament's strongest run differential despite finishing with 6 points.
- **Total Runs by Venue (Area/Line Chart):** An area chart was selected to show the distribution and relative magnitude of run-scoring across venues simultaneously. Hobart (2.3K) and Adelaide (2.0K) emerged as the highest-scoring grounds, indicating flat, batter-friendly pitches. Melbourne (1.1K) produced the fewest runs, suggesting conditions favouring bowlers — directly actionable intelligence for pitch-specific team selection.
- **Total Wickets by Venue (Bar Chart):** A descending bar chart allows immediate identification of the most wicket-productive venues. Hobart (100 wickets) and Sydney

(89) produced the most dismissals, consistent with their run totals and confirming high-scoring, high-wicket conditions as opposed to low-scoring, grinding contests.

- **Total Runs by Stages (Stacked Bar):** The horizontal stacked bar chart, segmented by Group Stage, Super 12, and Playoffs, visualises how deep each team progressed through the tournament. India (165 in Group Stage, 801 in Super 12) and Australia (577 in Super 12) demonstrate the run accumulation characteristic of teams playing knockout-stage cricket — of direct relevance to understanding performance under pressure.

5.2 Page 2 — Openers Analysis

Purpose

Openers set the tone of a T20 innings. Their ability to exploit powerplay restrictions (fielding circles in the first 6 overs), score at a high rate, and resist early wickets determines whether a team can access the full resource of 20 overs with wickets in hand. This page isolates batting positions 1 and 2 and evaluates performance across six critical KPIs.

KPI Ribbon

Total Runs	Batting Avg	Strike Rate	Boundary %	Ball Faced	Consistency
3,812	24.1	118.8	56.8	3,210	100.7

Visual Design Rationale

- **Scatter Chart — Batting Performance by Team (Batting Avg vs Strike Rate):** A scatter plot was selected over a bar chart because it reveals the two-dimensional trade-off between average (quality) and strike rate (aggression) simultaneously. Teams in the upper-right quadrant — those with both high averages AND high strike rates — represent the most complete opening partnerships. England and New Zealand openers occupied this elite quadrant, confirming their status as the most dangerous powerplay attackers in the tournament.
- **Top 5 Players by Highest Runs (Bar Chart):** Max O'Dowd (Netherlands, 242 runs) led all openers, followed by Jos Buttler (England, 225), Kusal Mendis (Sri Lanka, 223), Pathum Nissanka (Sri Lanka, 214), and Alex Hales (England, 212). This bar chart provides an unambiguous run-volume ranking and reveals that the top five featured players from five different teams, reinforcing that powerplay excellence is distributed across the tournament rather than concentrated in a single nation.

- **Detailed Player Table:** The scrollable player table exposes match-level records for every opener — enabling selectors to cross-reference aggregate KPIs against individual game context (opponent, venue, batting style). The conditional highlighting of Strike Rate above 150 immediately flags exceptional powerplay performances for further review.

Key Insight: Openers averaged a Boundary % of 56.8%, meaning more than half of all opener runs came from boundaries. This confirms the T20 powerplay as a fundamentally boundary-dependent phase and validates selection priority for players with proven boundary-hitting capability against new-ball bowling.

5.3 Page 3 — Middle Order Analysis

Purpose

The middle order (positions 3–6) is the structural backbone of a T20 innings. These batters must adapt to match situation — accelerating from a strong base or rebuilding from early collapses — while maintaining sufficient strike rate to set or chase competitive totals. The Consistency Index becomes the defining KPI on this page.

KPI Ribbon

Total Runs	Batting Avg	Strike Rate	Boundary %	Ball Faced	Consistency
5,102	24.6	121.7	48.1	4,192	105.4

Visual Design Rationale

- **Consistency Index by Player (Horizontal Bar Chart):** The Consistency Index — computed as (Average Runs / Standard Deviation of Runs) * 100 — is the signature metric of this page. Players with very high averages but high variance (one 80-run knock, three single-digit scores) will score lower than players with moderate but reliable returns. Calum MacLeod (Scotland, 2400 index) leads by a significant margin, reflecting his remarkable run stability at position 3. Evin Lewis (1400) and Gerhard Erasmus (900) follow, indicating middle-order reliability across diverse opposition.
- **Batting Performance by Team (Scatter Chart):** The strike rate vs. batting average scatter for middle-order players reveals Sri Lanka's middle order as the standout combination — high strike rate AND high average. Pakistan's middle order, by contrast,

shows a lower strike rate despite reasonable averages, suggesting a more conservative approach that may not be optimal in high-pressure T20 situations.

- **Player Detail Table:** The Boundary % column within the detail table enables selectors to distinguish between players who score through running and those who rely on boundaries — an important consideration when selecting against strong fielding sides that restrict running between wickets.

Key Insight: The middle order generated 5,102 total runs at an average of 24.6 — marginally higher than openers (24.1). The lower Boundary % (48.1% vs. 56.8% for openers) confirms that middle-order batters rely more on running and placement, making fitness and running ability a legitimate selection criterion for this batting phase.

5.4 Page 4 — All Rounders Analysis

Purpose

All Rounders represent the highest-value selection category in T20 cricket. A player who contributes meaningfully with both bat and ball effectively provides the selection flexibility of two specialists in one squad slot. This page applies a dual-dimensional evaluation framework that no traditional scorecard format accommodates.

KPI Ribbon

Total Runs	Batting Avg	Strike Rate	Wickets	Bowling Avg	Bowl SR
1,848	12.9	108.6	515	22.5	18.4

Visual Design Rationale

- **Most Reliable All Rounder (Scatter Chart — Strike Rate vs Bowling Economy):** A scatter chart was selected because it simultaneously encodes batting aggressiveness (X-axis: Strike Rate) and bowling containment (Y-axis: Bowling Economy, where lower is better). Players in the upper-right quadrant score quickly but concede runs; the ideal All Rounder occupies the lower-right (high Strike Rate, low Economy). This quadrant analysis immediately surfaces player profiles that balance both disciplines.
- **All Round Average Comparison (Clustered Bar + Line):** The combination chart showing Batting Average, Bowling Average, and Bowling Economy for the Top 5 All Rounders (Rashid Khan, David Wiese, Shadab Khan, JJ Smit, Jan Frylinck) allows

simultaneous comparison of all three dimensions. Rashid Khan's profile (Batting Avg 29, Bowling Avg 19, Economy 6.4) demonstrates the most balanced all-round contribution of the tournament.

- **Detailed All Rounder Table:** The table includes both batting columns (Runs, Boundary %) and bowling columns (Wickets, Dot Ball %, Economy) in a unified row per player per match, enabling game-level All Rounder impact assessment unavailable in any standard scorecard platform.

Key Insight: Rashid Khan (Afghanistan) delivered the tournament's most complete All Rounder performance — a Batting Average of 29 combined with a Bowling Economy of 6.4 and 19 Bowling Average. No other player in the tournament matched this combination, reinforcing Afghanistan's strategic dependence on a single multi-dimensional asset and the competitive disadvantage that creates.

5.5 Page 5 — Bowlers Analysis

Purpose

Bowling attack selection in T20 cricket involves resolving a fundamental tension: maximising wicket-taking potential while minimising run concession. These objectives are frequently in conflict — aggressive wicket-seeking bowlers often produce inconsistent economy rates. This page provides the analytical framework to identify bowlers who resolve this tension most effectively.

KPI Ribbon

Balls Bowled	Wickets	Economy	Bowl Avg	Bowl SR	Dot Ball %
9,494	515	7.3	22.5	18.4	40.3

Visual Design Rationale

- **Bowling Economy & Wickets by Bowler (Grouped Bar):** A grouped bar chart was selected to compare economy and wickets on the same visual without conflating the two axes. Nasum Ahmed (Bangladesh) achieved the tournament's best economy of 3.5 runs per over — an exceptional containment performance in a high-scoring tournament where the overall average was 7.3. Jason Holder (West Indies) combined 5 wickets with a 4.7 economy, identifying him as the most complete bowler in the chart.

- **Bowling Average & Wickets by Bowler (Dual-Line Chart):** A dual-line chart tracks both wickets taken and bowling average for the Top 5 by average. Anrich Nortje (South Africa) leads with 11 wickets at an average of 8.5 — the most wickets among the lowest-average bowlers, a combination that signals elite pace-bowling efficiency under tournament conditions.
- **Dot Ball % (Detail Table):** The inclusion of Dot Ball % in the bowler table is a deliberate analytical choice. In T20 cricket, dot balls build pressure that triggers rash batting shots — they are the indirect precursor to wickets. Bowlers with high dot-ball percentages create match-winning pressure even in spells where they do not directly take wickets. A tournament-average of 40.3% confirms that two in every five deliveries produced no run — a structurally significant pressure baseline.

Key Insight: Wanindu Hasaranga (Sri Lanka) took the most wickets in the tournament (15) — two more than any other bowler. His combination of leg-spin variety and aggressive wicket-seeking strategy represents the archetype of the T20 impact bowler: not the most economical, but the most match-defining.

5.6 Page 6 — Stats Leaderboard

Purpose

The Stats page consolidates the tournament's six most consequential performance rankings into a single, grid-format visual reference. It functions as both a quick-reference summary for selectors and a presentation-ready leaderboard for broadcast and communications use.

Panel	Metric	Top Performer	Value
Top 5 Batters by Runs	Total Runs	Virat Kohli (India)	296
Top 5 Highest Scores	Single Innings	Rilee Rossouw (SA)	109
Top 5 Batters by Avg	Batting Average	Virat Kohli (India)	99.0
Top 5 Bowlers by Wickets	Wickets	W. Hasaranga (SL)	15
Top 5 Bowlers by Economy	Economy Rate	Nasum Ahmed (BAN)	3.5
Top 5 Bowlers by Bowling Avg	Bowling Average	Glenn Maxwell (AUS)	6.3

The deliberate use of six independent bar charts — each with consistent teal colour theming and data labels — ensures visual parity across all panels. No single chart overwhelms the others, reinforcing the equal analytical weight of batting and bowling contributions to tournament success.

6. Business Recommendations & Conclusion

6.1 Squad Construction Framework

Based on the analytical findings presented across all six dashboard pages, the following framework is recommended for selectors constructing squads for future ICC T20 tournaments:

Opening Pair Selection Criteria

- **Primary Metric:** Prioritise Strike Rate above 130 combined with Boundary % above 55%. Openers who score primarily through running (below 45% boundary contribution) create insufficient pressure during the powerplay restriction window.
- **Secondary Metric:** Consistency Index above 80. An opener with a single 90-run performance and four single-digit scores is a selection liability — the variance creates match-outcome dependency on a single innings.
- **Benchmark:** Jos Buttler (England) and Alex Hales achieved a combined powerplay contribution that ranked first in the tournament. Their profile — right-right hand combination, high strike rate (163+), boundary-dominant scoring — represents the optimal opener archetype.

Middle Order Selection Criteria

- **Primary Metric:** Batting Average above 25 with Strike Rate above 120 in the Super 12 and Playoffs stages specifically (use the stage slicer to isolate these records).
- **Secondary Metric:** Demonstrated performance against both pace and spin bowling styles (Batting Style cross-filter in the player table enables this analysis).
- **Benchmark:** Virat Kohli's tournament-wide Batting Average of 99 across all middle-order innings represents the defining standard — consistent, high-average performance with sufficient strike rate to remain relevant across all match phases.

Bowling Attack Composition

- **Pace-Spin Balance:** The tournament data confirms a 60-40 pace-to-spin wicket distribution in Super 12 and Playoffs stages. Squads deploying fewer than two quality spinners face structural gaps against teams with strong middle-order bat-pad players.
- **Economy Floor:** No bowler with an economy rate above 9.0 in Super 12 matches should be selected for knockout cricket. The pressure accumulation at this level consistently produces single-over capitulations that are match-defining.

- **Dot Ball % Threshold:** Target a minimum of 38% dot ball rate across the bowling attack. The tournament average of 40.3% confirms this as an achievable standard; attacks falling below 35% consistently concede match-winning totals.

All Rounder Priority

- **Minimum Qualification:** An All Rounder must contribute meaningfully in both disciplines — defined as Batting Average above 15 AND Bowling Economy below 8.0 in Super 12 stage. Players meeting only one criterion should be classified as specialists.
- **Strategic Value:** The dashboard's All Rounder page confirms that squads with two qualified All Rounders (Rashid Khan profile) have structural batting depth AND bowling flexibility unavailable to specialist-heavy squads. This combination proved decisive in the knockout stages.

6.2 Venue-Specific Strategy Recommendations

The Venue analysis on the Overview page provides actionable pitch-specific intelligence:

Venue	Runs (Total)	Wickets	Pitch Profile	Selection Implication
Hobart	2,300+	100	Flat, high-scoring	Prioritise batting depth; play extra batter
Adelaide	2,000+	89	Good pace & carry	Genuine fast bowlers essential
Sydney	1,500	81	Balanced	Standard XI — no specialist adjustment required
Melbourne	1,100	49	Seam-friendly	Play 4 pace bowlers; reduce spin by 1
Brisbane	1,200	62	Hard, bouncy	Short-pitch specialists add value
Perth	1,200	55	Fast, bouncy	Favour aggressive pace over economy spin
Geelong	1,500	79	Spin-friendly	Premium leg-spinner is a match-winning asset

6.3 Stage-Specific Selection Intelligence

The stage slicer reveals a consistent pattern across the tournament: player performance rankings shift significantly between Group Stage and Super 12/Playoffs. Selectors should apply the following protocol:

1. Filter the batting and bowling pages to Super 12 and Playoffs stages exclusively when shortlisting players for knockout tournament squads.
2. Identify players whose Consistency Index improves (rather than declines) from Group Stage to Super 12 — these players demonstrate the ability to elevate performance under increased competitive pressure.
3. Discard Group Stage averages from selection discussions unless no Super 12 data is available for the player in question.

6.4 Conclusion

The ICC Men's T20 World Cup 2022 Analytics Intelligence Dashboard represents a significant advancement in the analytical infrastructure available to cricket selection committees. By consolidating batting, bowling, and match metadata into a unified, filter-responsive Power BI environment — supported by a robust Star Schema data model, 30+ bespoke DAX measures, and six role-specific analytical pages — the platform eliminates the information fragmentation that has historically characterised T20 selection decision-making.

The findings from this project support three overarching conclusions:

- **Conclusion 1 — Consistency Trumps Peak Performance:** Players with high Consistency Index scores outperform high-average, high-variance players in knockout cricket, where a single low-scoring innings cannot be recovered across a multi-game series.
- **Conclusion 2 — Venue Intelligence is Underutilised:** The runs-by-venue and wickets-by-venue data reveal significant ground-to-ground variation that traditional selection panels do not systematically account for. The Melbourne vs. Hobart differential alone justifies adjusting batting depth and bowling attack composition between fixtures.
- **Conclusion 3 — All Rounders Define Knockout Success:** The teams that progressed deepest into the Playoffs — England, India, New Zealand — all deployed multiple qualified All Rounders. The dataset confirms this is not coincidental; it reflects the structural advantage of batting-bowling flexibility in high-pressure, limited-over cricket.

The recommendations presented in Section 6.1 through 6.3 are directly grounded in the dashboard's analytical outputs and are intended to be revisited each tournament cycle as new data populates the model. As the dataset is extended to future ICC T20 World Cup editions, the framework will become progressively more powerful — moving from single-tournament pattern recognition to multi-year predictive capability.

It is the recommendation of this report that the Analytics Intelligence Dashboard be adopted as a standard pre-tournament preparation tool for all ICC Member Board selection committees, with particular emphasis on the stage-filtered Consistency Index analysis and the venue-specific batting depth recommendations outlined herein.

7. Appendix — DAX Measures Reference

The following table provides a complete reference index of all DAX measures implemented in the Power BI model. For the full code implementation of each measure, refer to the accompanying document: DAX_Measures_and_Calculated_Columns.pdf.

#	Measure Name	Category	Output Type
1	Total Runs	Batting	Integer
2	Average Runs	Batting	Decimal
3	Avg Ball Faced	Batting	Decimal
4	Avg Run Rate	Batting	Decimal
5	Batting Average	Batting	Decimal
6	Batting Position	Batting	Integer
7	Boundary %	Batting	Percentage
8	Consistency Index	Batting	Decimal
9	Highest Individual Score	Batting	Integer
10	StdDev Runs	Batting	Decimal
11	Strike Rate	Batting	Decimal
12	Total Ball Faced	Batting	Integer
13	Total Innings Batted	Batting	Integer
14	Total Innings Dismissed	Batting	Integer
15	Running Between Wickets	Batting	Integer
16	Runs from 4s	Batting	Integer
17	Runs from 6s	Batting	Integer
18	Total Overs Decimal	Batting	Decimal
19	Balls Bowled	Bowling	Integer
20	Bowling Average	Bowling	Decimal
21	Bowling Economy	Bowling	Decimal
22	Bowling Strike Rate	Bowling	Decimal
23	Dot Ball %	Bowling	Percentage
24	Run Conceded	Bowling	Integer
25	Top 5 Wickets	Bowling	Integer
26	Total Innings Bowled	Bowling	Integer
27	Wickets	Bowling	Integer
28	M (Matches)	Point Table	Integer

#	Measure Name	Category	Output Type
29	W (Wins)	Point Table	Integer
30	L (Losses)	Point Table	Integer
31	Pts (Points)	Point Table	Integer
32	NRR	Point Table	Decimal
33	Winner	Others	Text
34	Runner Up	Others	Text
35	Total Players	Others	Integer

End of Report

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